Hodge classes on six-dimensional complex abelian varieties: a first-method classification

Research draft (verified statements and explicitly conditional computations)

5 September 2026

Abstract

We organize the codimension-three Hodge problem for complex abelian sixfolds by combining the low-dimensional results of Moonen–Zarhin, the recent secant-sheaf theorem of Markman, Albert's classification, and isogeny decompositions. The K-Weil construction is placed in the correct degree: for $[K:\mathbf{Q}]=2$ and $\dim_K H^1(A,\mathbf{Q})=2n$, one has $W_K=\bigwedge_K^{2n} H^1(A,\mathbf{Q})\subset H^{2n}(A,\mathbf{Q})$; the signature (n,n) is required for these classes to be Hodge. A generic $SU(3,3)$ calculation gives $B^3=\mathbf{Q}\theta^3\oplus W_K$. Under an explicitly stated split Hodge-group hypothesis, we give Kunneth bases of sizes 5 and 8 for $S_2\times W_4$ and $E_1\times E_2\times W_4$. Shared CM fields and repeated factors invalidate that tensor formula and can create mixed or diagonal invariants. Markman's theorem covers the split-discriminant sixfold case only. The general six-dimensional Hodge conjecture therefore remains open.

1 Hodge classes, Weil type, and isogeny reduction

Let $A/\mathbf{C}$ be an abelian variety of dimension g. We write

$$B^p(A):=H^{2p}(A,\mathbf{Q})\cap H^{p,p}(A)$$

for rational Hodge classes. Equivalently, with the Tate twist explicit, $B^p(A)=H^{2p}(A,\mathbf{Q}(p))^{\mathrm{MT}(A)}=H^{2p}(A,\mathbf{Q})^{\mathrm{Hg}(A)}$, where $\mathrm{Hg}(A)$ is the Hodge group; see Milne [5, Prop. 1.1]. Let $D^\bullet(A)\subset B^\bullet(A)$ denote the subalgebra generated by divisor classes, so $D^1(A)=\mathrm{NS}(A)_{\mathbf{Q}}$.

1.1 The K-Weil space

Suppose that an imaginary quadratic field K acts on $H^1(A,\mathbf{Q})$ and $\dim_K H^1(A,\mathbf{Q})=2n$ (thus $\dim A=2n$). The K-linear exterior line and its underlying rational space are

$$W_K(A):=\bigwedge_K^{2n} H^1(A,\mathbf{Q}),\qquad \dim_{\mathbf{Q}} W_K(A)=2,$$

with the natural inclusion

$$W_K(A)\subset\bigwedge_{\mathbf{Q}}^{2n} H^1(A,\mathbf{Q})=H^{2n}(A,\mathbf{Q}).$$

The inclusion $W_K(A)\subset B^n(A)$ is an additional Hodge condition, not a consequence of the K-action alone. For an imaginary quadratic K, it holds for a polarized Weil-type structure when the two K-eigenspaces in $H^{1,0}(A)$ have equal dimension (n,n) (the Weil signature hypothesis). We use W_K only for this rational two-dimensional space and explicitly state when it is known to consist of Hodge classes.

Proposition 1.1 (Isogeny reduction)**.** *If $f : A \to A'$ is an isogeny, then f^* identifies $B^p(A')$ with $B^p(A)$ and preserves algebraicity of classes. Hence one may replace A by an isogenous product $\prod_i S_i^{m_i}$ of powers of simple factors (Poincare complete reducibility).*

Proof. The map f^* is an isomorphism on rational cohomology and a morphism of Hodge structures; $f_* f^* = (\deg f)\,\mathrm{id}$. Pullback and pushforward of algebraic cycles are algebraic. □

Proposition 1.2 (Split Hodge-group Kunneth formula)**.** *Let $A = \prod_{i=1}^r A_i$. Assume as a hypothesis that $\mathrm{Hg}(A) = \prod_i \mathrm{Hg}(A_i)$, or more generally that the derived Hodge groups are independent and the only common factor is the central weight torus. Then*

$$B^p(A) = \bigoplus_{p_1+\cdots+p_r=p} \bigotimes_i B^{p_i}(A_i).$$

This is a sufficient splitting hypothesis for the calculation, not a claim that the full Mumford–Tate groups form a direct product. Without it, isogenous factors, common CM fields, and nonzero Hom*-classes may enlarge the invariant space.*

2 Fourfold benchmark and Markman's theorem

Theorem 2.1 (Moonen–Zarhin, quoted)**.** *For complex abelian varieties of dimension at most five, Moonen–Zarhin describe the Hodge algebra in terms of divisor classes, Weil classes, and (in the relevant Albert types) exceptional tensors; see [1, 2]. In particular, for a generic imaginary-quadratic Weil fourfold W_4 with $\mathrm{rk}\,\mathrm{NS}(W_4) = 1$ and $\dim_{\mathbf{Q}} W_K = 2$,*

$$B^0 = \mathbf{Q}, \quad B^1 = \mathbf{Q}\theta, \quad B^2 = \mathbf{Q}\theta^2 \oplus W_K, \quad \dim_{\mathbf{Q}} B^3 = 1,$$

where θ is an ample divisor class and $L^2 : B^1 \xrightarrow{\sim} B^3$ is Hard Lefschetz. Thus

$$P_{W_4}(t) = 1 + t + 3t^2 + t^3 + t^4.$$

The assertion about the Hodge algebra is a structural theorem; algebraicity of Weil classes requires the additional results quoted below.

Theorem 2.2 (Markman, quoted; Theorem 1.5.1)**.** *Let A be an abelian sixfold of imaginary-quadratic K-Weil type and let $[\delta] \in \mathbf{Q}^*/N_{K/\mathbf{Q}}(K^*)$ be the polarization discriminant class. Markman proves algebraicity of the two-dimensional Weil space $W_K \subset H^6(A, \mathbf{Q})$ when*

$$[\delta] = [-1] \quad \text{in } \mathbf{Q}^*/N_{K/\mathbf{Q}}(K^*).$$

The same work obtains a fourfold Weil-class consequence for arbitrary discriminant by degeneration [3]. No claim here extends the sixfold theorem to non-split discriminant classes.

3 Simple sixfolds: Albert-type schematic baseline (first method)

Let $D = \mathrm{End}^0(A)$, let E_0 be the maximal totally real subfield of the center of D, and write $e = [E_0 : \mathbf{Q}]$ and m_{rel} for the relative multiplicity. The following table is a schematic generic PEL baseline, not a complete parameter classification of all Albert IV cases and not the full Hodge algebra on special Hodge loci.

Albert type	Constraint for $g = 6$	Generic PEL group and divisor Hilbert series	$[t^3]$
I	$g = em_{\mathrm{rel}}$; $(e, m_{\mathrm{rel}}) = (1,6), (2,3), (3,2), (6,1)$	$\mathrm{Res}_{E_0/\mathbf{Q}} \mathrm{Sp}_{2m_{\mathrm{rel}}}$; $P_D(t) = (1 + t + \cdots + t^{m_{\mathrm{rel}}})^e$	$1, 4, 7, 20$
II	$g = 2em_{\mathrm{rel}}$; $(e, m_{\mathrm{rel}}) = (1,3), (3,1)$	Quaternionic PEL group; invariant ring must be computed case by case	not determined
III	same numerical possibilities as II	Quaternionic PEL group; Moonen–Zarhin Criterion 13 gives an exceptional-tensor occurrence criterion	not determined
IV	CM centre, Schur index and relative multiplicity subject to the standard Albert dimension relation $g = 6$	Unitary/CM groups; imaginary-quadratic K-Weil type is only a subfamily; full parameters omitted here	not determined

Remark 3.1. *For Type I, the polynomial in the table is a* generic PEL invariant/divisor *computation. Special Hodge loci may have smaller Hodge group and additional classes. Types II and III require a six-dimensional, parameter-by-parameter invariant and algebraicity analysis. Criterion 13 of [2] concerns occurrence of exceptional tensors; it is not an algebraicity theorem. Type IV is displayed only schematically: a complete list would require the standard center, Schur-index, and relative-multiplicity parameters, which is beyond this draft.*

4 A conditional generic $SU(3,3)$ computation

Assume that A is a sixfold of imaginary-quadratic K-Weil type with signature $(3,3)$, generic Mumford–Tate group, and $\mathrm{rk}\,\mathrm{NS}(A) = 1$. After base change to $\mathbf{C}$ and removal of the central similitude, assume the derived Hodge group is $\mathrm{SL}(U)$ with $\dim_{\mathbf{C}} U = 6$ and $H^1(A, \mathbf{C}) = U \oplus U^\vee$. Rational invariants are obtained by Galois conjugating the two determinant lines.

Proposition 4.1 (Representation computation; conditional)**.** *Under the preceding hypotheses,*

$$\left(\bigwedge^6 (U \oplus U^\vee)\right)^{\mathrm{SL}(U)} = \det U \ \oplus \ \det U^\vee \ \oplus \ \mathbf{C}\, c,$$

where c is the scalar contraction in $\bigwedge^3 U \otimes \bigwedge^3 U^\vee = \mathrm{End}(\bigwedge^3 U)$. The conjugate determinant lines form W_K, while c corresponds to θ^3. Consequently

$$B^3(A) = \mathbf{Q}\theta^3 \oplus W_K, \qquad \dim_{\mathbf{Q}} B^3(A) = 3.$$

Proof. Decompose $\bigwedge^6 (U \oplus U^\vee) = \bigoplus_{a+b=6} \bigwedge^a U \otimes \bigwedge^b U^\vee$. For $a = 0, 6$ one obtains the determinant lines. For $0 < a < 6$, irreducibility leaves an invariant only for $a = b = 3$, where Schur's lemma gives the scalar endomorphism. This is a computation for the stated derived group, not a classification of all Type-IV sixfolds. □

5 Non-simple products and explicit Kunneth bases

For a generic simple d-fold with divisor-generated Hodge algebra and rank-one Néron–Severi group, write $P_d(t) = 1 + t + \cdots + t^d$. The following baseline table is conditional on that hypothesis for

each factor and on split Hodge groups (with only the common weight torus allowed).

decomposition	Hilbert product	$\dim B^3$
$1+5$	$(1+t)(1+t+\cdots+t^5)$	2
$2+4$	$(1+t+t^2)(1+t+\cdots+t^4)$	3
$3+3$	$(1+t+t^2+t^3)^2$	4
$1+1+4$	$(1+t)^2(1+t+\cdots+t^4)$	4
$1+2+3$	$(1+t)(1+t+t^2)(1+t+t^2+t^3)$	6
$2+2+2$	$(1+t+t^2)^3$	7

These numbers are not asserted when factors are isogenous or share CM data.

5.1 $S_2 \times W_4$: a five-element basis

Let S_2 be a generic simple abelian surface with $B^\bullet(S_2) = \mathbf{Q}[s]/(s^3)$, and let W_4 satisfy the fourfold hypotheses above. Assume the Hodge groups split, $\mathrm{Hg}(S_2 \times W_4) = \mathrm{Hg}(S_2) \times \mathrm{Hg}(W_4)$ (with the common weight torus handled by the Tate twist). For a rational basis u_1, u_2 of $W_K \subset B^2(W_4)$,

$$B^3(S_2 \times W_4) = \langle w^3,\ sw^2,\ su_1,\ su_2,\ s^2w \rangle_{\mathbf{Q}},$$

where $w = \theta_{W_4}$. The first, second and fifth classes are divisor-generated; su_1, su_2 are products of divisors with Weil classes. Hence $\dim_{\mathbf{Q}} B^3 = 5$. Their algebraicity follows when u_i are algebraic (for example in the fourfold range covered by Markman's consequence).

5.2 $E_1 \times E_2 \times W_4$: an eight-element basis

Let E_1, E_2 be non-isogenous elliptic curves, and assume the full split Hodge-group condition $\mathrm{Hg}(E_1 \times E_2 \times W_4) = \mathrm{Hg}(E_1) \times \mathrm{Hg}(E_2) \times \mathrm{Hg}(W_4)$ (with only the common weight torus shared). With $e_i \in B^1(E_i)$ and $w = \theta_{W_4}$,

$$B^3 = \langle w^3, e_1w^2, e_2w^2, e_1e_2w, e_1u_1, e_1u_2, e_2u_1, e_2u_2 \rangle_{\mathbf{Q}},$$

so $\dim_{\mathbf{Q}} B^3 = 8$. The first four classes are divisor-generated and the last four are mixed products with Weil classes. If the elliptic factors are isogenous or share a CM field, diagonal/mixed invariants appear and this basis is not asserted.

6 Status of classes and open questions

- **Theorem (quoted):** divisor classes are algebraic; Moonen–Zarhin give low-dimensional structural results; Markman proves sixfold K-Weil algebraicity under $[\delta] = [-1]$ and the stated Weil hypotheses.
- **Computation (conditional):** the generic $SU(3,3)$ invariant calculation gives three codimension-three lines; the split-group Kunneth calculations give dimensions 5 and 8 with the displayed bases.
- **Open:** non-split sixfold discriminants; complete Albert II/III and non-quadratic IV sixfold invariants and algebraicity; special CM/unitary loci; mixed-Weil classes for shared CM fields; diagonal invariants and correspondences for repeated factors; and the secant-sheaf ch_3 determinant at Markman's special point $J(C) \times \widehat{J(C)}$.

No result in this paper proves the Hodge conjecture for a general complex abelian sixfold. In particular, the general six-dimensional case remains open.

References

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[2] B. J. J. Moonen and Yu. G. Zarhin, "Weil classes on abelian varieties," preprint (1996), Criteria 4 and 13. https://arxiv.org/abs/alg-geom/9612017.

[3] E. Markman, "Cycles on abelian $2n$-folds of Weil type from secant sheaves on abelian n-folds," arXiv:2502.03415 (2025). https://arxiv.org/abs/2502.03415.

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